RDIMS#1184211

ROYAL CANADIAN NAVY
QUALIFICATION STANDARD AND PLAN (QSP)

Fire Control Supervisor (FCS)

Naval Electronic Sensor Operator (NES OP) - 00115

MITE ID – 122543

NQUAL – ALBA

NPTG HQ Approval -07/06/2021
Version No: 1.9
MITE Crse: 122543

RECORD OF AMENDMENTS

The training authority (TA) designated for this qualification is the Commander Naval Personnel and Training Group (CNPTG). Naval Training Development Centre Atlantic (NTDC(A)) is the designated development centre responsible for the maintenance of training. Amendments to PO standard are to be requested through NTDC(A) and approved by CNPTG. Ammendments to Lesson Guidence are to be approved by CO NTDC(A).

Revision	Approval Date	Authority
1.9	07 Jun 2021	NPTG HQ

TABLE OF CONTENTS

Title		Page
FOREWORD		3
CHAPTER 1 : TRAINING MANAGEMENT DETAILS		4
	Aim	5
	Course Duration	5
	Course Capacity	5
	Prerequisites	6
	Prior Learning Assessment and Recognition (PLAR)	6
	Resources	6
	Special Instructions	6
CHAPTER 2 : PERFORMANCE OBJECTIVES		7
PO 001	Lead FC Team Preparations	8
EO 001.01.01	Pre-Firing Preparations	9
EO 001.01.01	Supervise MASS Loading Procedures	11
EC 001.01	Supervise Pre-Firing Preparations	12
EO 001.02 .01	PAC Theory	14
EO 001.02.02	Confirming System settings	15
EC 001.02	Optimize FC Settings	17
PC 001	Lead the FC Team Preparations	20
PO 002	Supervise FC Drills	23
EO 002.01.01	ESSM Air Engagements	24
EO 002.01.02	ESSM Misfire	25
EC 002.01	Supervise ESSM Engagement	26
EO 002.02.01	PAC	28
EC 002.02.01	Supervise PAC	29
EO 002.02.02	57mm Air Primary	31
EO 002.02.03	57mm Surface procedures	32
EO 002.02.04	Optronic Secondary	34
EO 002.02.05	Warning Shots	35
EO 002.02.06	57mm Misfire	36
EC 002.02.02	Supervise 57mm Engagement	37
EO 002.03.01	CIWS AWW Manual	40
EO 002.03.02	CIWS Surface	41
EC 002.03	Supervise CIWS Engagement	42
EO 002.04.01	System troubleshooting	44
EO 002.04.02	Advising AWWD on System	45
EO 002.04.03	Determine Faults and advise impact	46
EC 002.04	Advise AWWD on functionally of FC systems	47
PO 002	Consolidation	49
PC 002	Supervise FC drills	50
CHAPTER 3: ASSESSMENT OF TRAINEE		54
ANNEX A	TASK LIST	A-1
ANNEX B	QSP WORKING GROUP RECORD OF PROCEEDINGS	B-1

ANNEX C	COSTING INFORMATION	C-1
ANNEX D	ASSESSMENT PLAN	D-1
ANNEX E	SCHEDULING INFORMATION	E-1
ANNEX F	TEL DEVELOPMENT	F-1
ANNEX G	REFERENCES	G-1
ANNEX H	PLAN, TRAIN, DO	H-1

FOREWORD

1. This training is based on the specification found at:

   a. A-PD-055-002/PP-002 NESOP MOSID 00115 Occupation Specifications dated 06 May 2019.

2. A trainee will achieve the qualification **ALBA** when the standards listed in all Performance Objectives have been met.

3. The designated TEs for this training is indicated below:

TE	Phase / Module	MITE Course ID	NQual
NFS(A)/NFS(P)	All	122543	ALBA

CHAPTER 1: TRAINING MANAGEMENT DETAILS

AIM

1. The aim of this Qualification Standard and Plan (QSP) is to establish the requirements and training for the RQ MS FCSup course. This QSP is designed to assist NTDC(A) in developing, Naval Fleet Schools in conducting and evaluating, and NPTG in validating the training.

COURSE DURATION

2. Course duration is calculated in accordance with (IAW) NPTORD 4500-22, Annex B. For this course, a period of instruction, administration and PT is 50 minutes in duration. The recommended schedule is based on **maximum load**. Scheduling cells are to increase/decrease course length based on number of students enrolled on the course.

Course Duration Calculation	
Allocation of ATT	**Periods**
Sum of all EOs/POs	49
Enabling Checks (ECs)	40
Performance Checks (PCs)	17
Actual Training Time (ATT) Sub-total	**14 trg days**
Allocation of BAT, TAT, TTT and PT IAW NPTORD 4500-22, Annex B Table 3.1	
Base Administrative Time (BAT)	5
Training Administrative Time (TAT)	5
Training Travel Time (TTT)	0
Physical Fitness (PT)	9
Sub-total	3 **trg days**
Course Duration Calculation (ATT+BAT+TTT+TAT+PT) **1400 minutes** * **Partial days must be rounded up**	17 **trg days**

COURSE CAPACITY

3. Planning for the conduct of training is based on the guidelines in NPTORD 4500-40.

   a. Maximum – 12;

   b. Minimum – 3; and

   c. Optimum – 6.

PREREQUISITES

4. Prerequisites. To be selected for this course, the loading authority must ensure that trainees have the following prerequisites:

a. Be a qualified Senior Electronic Warfare Analyst (ALAZ); and

b. Hold a security clearance of Level III.

PRIOR LEARNING ASSESSMENT AND RECOGNITION (PLAR)

PLARs shall be conducted IAW NPTORD 4500-27, DAOD 5031-1 and CFITES Vol. 12.

RESOURCES

DEDICATED COURSE STAFFING				
TITLE	RANK	QUALIFICATION	QTY	PO / EO
Course Officer	Lt(N)	AWWD (ALBU)	1	ALL
Instructor	MS	RQ-NESOP-MS (ALBA)	1 (2 for course load 6 or more)	ALL

GUEST LECTURERS			
AGENCY	RANK – MOSID	Lesson	TIME
CFMWC		001.02A	50 min

SPECIAL INSTRUCTIONS

Weapon Certification

5. Upon completion of the RQ MS Fire Control Supervisor course, the student will be considered Weapons Qualified for the purpose of carrying out the duties and fulfilling the role of Fire Control Supervisor onboard the ship. However, this does not preclude the individual from completing Weapons Certification with the AWW Team in his/her unit. The Fire Control Supervisor will still be required to certify with the AWW Team when Command deems it is necessary to do so.

CHAPTER 2: PERFORMANCE OBJECTIVES & LESSON GUIDANCE

PO 001 Lead Fire Control Team Preparations

1. **Performance:** Lead Fire Control team preparations

2. Conditions:

a. Given:
   (1) Fire Control team
   (2) Firing orders;
   (3) MASS brief Handouts;
   (4) Weather report/ Ballistic Data;
   (5) References;
   (6) Supervision

b. Denied: Nil.

c. Environmental: Alongside/At Sea ,Day/Night: Environment: Alongside/At Sea , Day/Night

3. **Standard**: The FC Sup will lead the Fire Control team preparations by:

a. Supervising Pre Firing Preparations to include:

   (1) Distributing Fire Orders;
   (2) Posting precautionary signage; and
   (3) Directing MASS loading (C00480); and

b. Optimizing FC settings by:

   (1) Confirming system settings by:
      i. Operating combat systems (CS) equipment (B00080); and
      ii. Supervising Fire Control (FC) system operations (D00040, D00130);
   (2) Supervising FC team tracking runs (D000110);
   (3) Assess Jamming (C00050); and
   (4) Carryout Electronic Protect (EP) measures (C00070).

EO 001.01 – Supervise Pre Firing Preparations

001.01.01 – Pre-Firing Preparations

Conditions		Sequence
Unclass Classroom		
Time	**Classification**	**Instructor**
100 min	Unclassified	RQ-MS Instructor

Learning Objectives	Method
1) Distribution of Firing Orders 2) Posting of Precautionary Signs 3) Striking Guardrails	Student Reading Card Sort PowerPoint/Video

Activity Steps

1. Students to read NAVORD 4995-0 page 1-15 (chapters 1-3) (paper copy if DWAN not available) (45 min)
2. Students are put in two groups. One group is to create a list of all the locations safety signage must be posted.
   Other group to write a "script" for briefing subordinates on safety precautions for striking guard rails.
   Students have 10 min to create their list/script. (10 min)
3. Students to present their lists/script to the class, instructor to provide feedback. (15 min)
4. Students are then split into 3 equal groups and provided cards to sort in order of "what order are the preparations conducted". Instructor to confirm/discuss order of duties groups selected. (15 min)
   Cards to include:
   a. Strike guard rails (upon slipping or prior to shoot)
   b. Distribute Firing Orders (48hrs prior)
   c. Post Precautionary Signs (prior to firing)
   d. Close up FC Team (prior to firing)
   e. Warning Pipes Commence (15 min prior)
   f. Team conducts FC tracking assessment (prior to firing)
   g. Commence Firing
      **** there will be some discussion on the order of duties, if students are able to justify their reasoning and it is sound, the class will discuss

5. Upon completion of first card sort (duties), second set of cards to be distributed to students to align with the appropriate step (5 min)
   Instructor to provide feedback to groups. (10)

Resources	References
1) Paper Copies of NAVORD 4995-0 2) Card Set of Duties (3 sets) 3) Card set for timings (3 sets) 4) Tape 5) PowerPoint on tracking assessment 6) Video on tracking assessments 7) CFMWC video on effect of MASS on FC tracking	1) NCPM 1000 2) NAVORD 4995-0

Notes to Instructor

Card Sort: Student are provided a set of cards (one duty per page) to put in order. For details on how to do a Card Sort see “slide sets” in AIT manual.
Cards to include titles A-G and second set to be the greyed out “response” following

EO 001.01 – Supervise Pre Firing Preparations

001.01.02 – Supervise MASS Loading Procedures

Conditions		Sequence
Unclass classroom		
Time	**Classification**	**Instructor**
100 min	Unclassified	RQ MS Instructor

Learning Objectives	Method
1) Direct MASS loading procedures (brief) 2) Manipulate MASS launch for loading 3) Supervise MASS loaders to confirm compliance in carrying out the load procedures	Practical Video Demo

Activity Steps	
	1. Instructor to provide handout that demonstrates MASS loading brief and have students read it aloud to the class. 2. Students to pair up and practice conducting the MASS loading brief. 3. Instructor to play demo video of MASS loading 4. Students to observe the MASS loading and indicate errors of the loading team WRT compliance in load procedures. (pause between scenarios) 5. Instructor to ask what items each student picked up when observing the various situations.

Resources	References
1) MASS loading brief handout 2) MASS loading demo videos	1) NCPM 1000 2) NAVORD 4995-0

Notes to Instructor

EC 001.01 – Supervise pre-firing preparations

Scope	Conditions	Sequence
EO 001.02	Instructor office, conference room, or empty classroom	After EO 001.02
Time	**Classification**	**Instructor**
150 min	Unclassified	RQ MS Instructor RQ PO2 Instructor (or above)

Scenario	
Assessment Steps	Input: Students are preparing for a live firing event and must brief their FC team Performance: Student must conduct their duties as FC Sup IAW checklist 001.02. (based on training scenario details determined by instructor ***see instructor notes for criteria) Output: IAW checklist 001.02 Feedback: immediate verbal and written IAW checklist 002.02A

Resources	References
1) Picture of the MASS loader panel	1) 2)

Notes to Instructor

Assistance shall not be provided to students during the assessment. Support manning IAW NWTP vol 2.

1) Firing scenario should be a two-weapon system shoot in any combination of ESSM, 57mm, or CIWS.
2) Pre-firing preparations scenario should begin with a hypothetical situation where the decided upon two-weapon system engagement. The FC Sup has to delegate and brief a novice OS on the responsibilities of distributing firing orders, posting precautionary signs (including listing their specific locations), and briefing the FC team on the striking of guardrails.
3) MASS scenario shall include the student giving a safety brief to the instructor followed by manipulation of the MASS control panel and confirming compliance of correct loading procedures iaw Closing Up and Clearing Away Procedures. The MASS control panel will be simulated using a provide picture of the loader panel. The student must clearly and accurately describe their exact manipulations of the system.

Limitations	
Impact of Failure	Feedback and additional instruction provided (when possible)

Item	EC 001.01 Supervise pre-firing preparations	SAT	UNSAT	Remarks
1. Pre-Firing Preparations				
Student correctly lists recipients of firing orders		☐	☐	
Student correctly lists locations for precautionary signs for given scenario		☐	☐	
Students brief covers safety as well as steps to ensure safe evolution		☐	☐	
2. Supervise MASS Loading Procedures				
Students brief covers safe dress, correct way to carry rounds and types of hazards associated with MASS		☐	☐	
Student correctly manipulates system ensuring safe evolution		☐	☐	
Student ensures compliance of correct loading procedures		☐	☐	

Score: /6

Trainee must achieve 5/6 SATs in order to pass this EC.

Positive Feedback	Improvement Feedback

Overall Assessment – Circle One			
Met Standard		**Did Not Meet Standard**	
Assessor I have discussed the results of this assessment with the candidate.		**Candidate** I have been debriefed on the result of this assessment	
Assessor	Name:	Date:	Signature:
Candidate	Name:	Date:	Signature:

EO 001.01 – Optimize FC Settings

001.02.01 – PAC Theory for Optimizing FC System

Conditions		Sequence
Unclass Classroom (optional with VTC capability)		
Time	**Classification**	**Instructor**
100 min	Unclassified	Guest Lecture: CFMWC or RQ-MS Instructor

Learning Objectives	Method
1) Function of PACs 2) How the gun records of Fall of Shot (FOS) Data	Think, Pair, Share Lecture Group modified "3 STAR"

Activity Steps	
	1. Instructor to post to class "what is the purpose of PACs?" Students are to think about their answer and write their response down. (5 min) Students are then asked to pair up and share their responses with their partner. (5 min) Instructor to then get groups to share their responses. (5 min) Anticipated Response: Method to used to calibrate weapons system to maximize kill ratio for incoming air contacts. 2. Guest Lecture or Instructor to provide a lecture on muzzle velocity, Fall of Shot Data and its significance in gunnery shoots. (45 min) 3. Students are asked to select one thing from the lesson that was new or particularly interesting to them and write it on the whiteboard. Instructor to discuss responses with class. (10 min).

Resources	References
1) Muzzle Velocity & FOS PowerPoint 2) VTC Facility	1) NCPM 1000 2) CEROS technical manual

Notes to Instructor
If guest lecture from Warfare Centre available provide courseware PowerPoint to ensure they do not provide information beyond the needs of students.

EO 001.02 – Optimize FC Settings

001.02.02 – Confirming System Settings

Conditions		Sequence
SEACOT (preferred) SEAWOLF/SETT (secondary option)		
Time	**Classification**	**Instructor**
50 min	Up to SECRET	RQ-MS Instructor or above

Learning Objectives	Method
1) Closing-Up and Clearing-Away (CUCA) 2) Confirm Meteorological Data 3) Confirm Ballistic Data 4) Confirm Policy Order is correct 5) Conduct FC Tracking Assessment	Collaborative problem solving Student developed CASE STUDY

Activity Steps	
	1. Ask students to list the steps of closing up and clearing away. Ask students who can grant them permission to train and elevate the mount. Ask students what could be the possible repercussions of moving the gun without permission. 2. The instructor to provide a copy of a weather report (received from MET Tech) to students. Students are to highlight the important information form the report and what details would be required to confirm/entered in the system. (5min) Anticipated Responses: a. sea state b. air temperature c. wind (direction and speed) d. check system 3. Instructor to pose question to class: “How do we confirm batch lot data?” (5 min) Anticipated Response: CSE will check batch lot on the rounds container down in the 57mm Magazine, then operator to input the data in the system for the initial shoot. ** Instructor to acknowledge that they may encounter resistance from Tech to confirm this data. 4. Instructor to break class into two teams, each with a flipchart. Then pose question to class “What information needs to be confirmed regarding policy order in the system?” Students are to then create lists from memory the information that must be confirmed based on policy. (5 min) Anticipated Response: a. FA b. PD

	c. MTI d. JAMMER 5. Instructor to then break students into groups of 2-3 (ensure students with less system experience are matched with ones with lots of experience). Students are to create their own CASE STUDY (based on either experience or completely fictional) where FC settings were not confirmed or inaccurate, detailing the scenario and the outcome of such failures. (15 min) 6. Groups to then present their CASE STUDY to the class, instructor to provide any feedback on the scenario and the actions if not covered. (20 min) 7. Instructor to then utilize either PowerPoint presentation (or video once developed) demonstrating smooth tracking or unsatisfactory tracking for a "tracking assessment". Instructor to reinforce the importance of tracking assessments and why we conduct them. (20 min)

Resources	References
1) Flip chart and markers 2) Met reports	1) NCPM 1000

Notes to Instructor

EC 001.02 – Optimize FC settings

Scope	Conditions	Sequence
EO 001.01	SEACOT (preferred) SEAWOLF/SETT (secondary option)	After EO001.01
Time	**Classification**	**Instructor**
200 min	Up to SECRET	RQ-MS

Scenario	
Assessment Steps	Input: Students provided a FC Operator during preparations for a live firing scenario. Students will close up their team and then be provided a MET tech report at the start of the run. The EC will conclude upon completion of tracking runs. Performance: Student must conduct their duties as FC Sup IAW checklist 001.01 Output: IAW checklist 001.01 Feedback: immediate verbal and written IAW checklist 001.01

Resources	References
1) Muzzle Velocity & FOS PowerPoint	1) NCPM 1000 2) CEROS Technical Manual

Notes to Instructor

Assistance shall not be provided to students during the assessment. Support manning IAW NWTP vol 2.

The scenario for the student will include a (Novice) FCO. The student must close-up their team in preparation for a PAC. Provide the student with MET Data and Ballistic Data. Once the FCS has verified the Met and ballistic settings, a policy report shall be issued over the net. The FCO should stumble through the process prompting the student to correct him.

This scenario must include one of each error of the following categories:

- Incorrect CUCA steps (not pairing each radar to the gun, not selecting TV power, not selecting radar power, etc)
- Incorrect MET data entry
- Incorrect ballistic data entry
- Incorrect system setup (IAW policy issued)

In the tracking run scenario, the instructor is to generate either a surface or air target into the system for the purpose of tracking and monitoring the A-Scope. The instructor will use a sticky-note to either show satisfactory or degraded tracking.

An incorrect voice procedure conducted by FC Op

Limitations	
Impact of Failure	Feedback and additional instruction provided (when possible)

Item	EC 001.02 Optimize FC settings	SAT	UNSAT	Remarks
1. Confirming System Settings				
QA CUCA process		☐	☐	
Student request permission to train and elevate the mount prior to bringing the gun on-line		☐	☐	**CRITICAL**
QA input of correct MET data		☐	☐	
QA input of correct ballistic data		☐	☐	
QA input of correct system setting IAW issued policy order		☐	☐	
Correctly assess tracking runs IAW NCPM 1000 and CEROS 200 Manual		☐	☐	

Score: /6

Trainee must achieve 5/6 SATs on non-critical factors **AND** all critical factors to pass this EC.

Positive Feedback	Improvement Feedback

Overall Assessment – Circle One			
Met Standard		**Did Not Meet Standard**	
Assessor I have discussed the results of this assessment with the candidate.		**Candidate** I have been debriefed on the result of this assessment	
Assessor	Name:	Date:	Signature:
Candidate	Name:	Date:	Signature:

PC 001 – Lead the FC Team Preparations

Scope	Conditions	Sequence
PO 001	SEACOT (preferred) SEAWOLF/SETT (secondary option)	After EC 001.02
Time	**Classification**	**Instructor**
250 min	Up to SECRET	RQ MS Instructor RQ PO2 Instructor (or above)

Scenario	
Assessment Steps	Input: Students provided a FC Operator for pre-firing preps and FC tracking runs. Performance: Student must conduct their duties as FC Sup IAW checklist 001.02 for a 20 min run. (based on training scenario details determined by instructor ***see instructor notes for criteria) Output: IAW checklist 001.02 Feedback: immediate verbal and written IAW checklist 002.02A

Resources	References
1) 2)	1) 2)

Notes to Instructor

1) Note for instructor: Pre-firing preparations scenario should begin with a hypothetical situation where the FC Sup has to delegate the responsibilities of distributing firing orders, posting precautionary signs, and the striking of guardrails to a novice OS.
2) MASS scenario shall be of a ship proceeding to sea and requiring a full load out including the student giving a safety brief to instructor followed by manipulation of the MASS control panel and confirming compliance of correct loading procedures.

Limitations	
Impact of Failure	Feedback and additional instruction provided (when possible)

Item	PC 001 Performance Statement	SAT	UNSAT	Remarks
1. Optimize FC Settings				
Have student explain purpose of PAC		☐	☐	
Have student explain importance of muzzle velocity		☐	☐	
		☐	☐	
2. Confirming System Settings				
Have student explain why entering correct weather patterns are important		☐	☐	
Have student explain why the particulars for each batch load are important		☐	☐	
		☐	☐	
3. Supervise Pre Firing Preparations				
Student correctly lists recipients of firing orders		☐	☐	
Student correctly lists locations for precautionary signs for given scenario		☐	☐	
Students brief covers safety as well as steps to ensure safe evolution		☐	☐	
4. Confirming System Settings				
Students brief covers safe dress, correct way to carry rounds and types of hazards associated with MASS		☐	☐	
Student correctly manipulates system ensuring safe evolution		☐	☐	
Student ensures compliance of correct loading procedures		☐	☐	

Score: /12

Trainee must achieve 10/12 SATs on non-critical factors.

Positive Feedback	Improvement Feedback

Overall Assessment – Circle One			
Met Standard		**Did Not Meet Standard**	
Assessor I have discussed the results of this assessment with the candidate.		**Candidate** I have been debriefed on the result of this assessment	
Assessor	Name:	Date:	Signature:
Candidate	Name:	Date:	Signature:

PROTECTED B (when complete)

PO 002 Supervise Fire Control Drills

1. Performance: Supervise Fire Control Drills (D00040, D00130)

2. **Conditions:**

a. Given:

(1) Fire Control team
(2) Firing orders;
(3) Supervision;
(4) MET report;
(5) Communication Terminal;
(6) CEROS display;
(7) FCO; and
(8) References.

b. Denied: Nil

c. Environment: Alongside/At Sea, Day/Night: Environment: Alongside/At Sea, Day/Night

c.

(1) Day or night

3. Standard: The FC Sup will supervise Fire Control drills (D00040, D00130) to include:

a. Supervising an ESSM engagement;

b. Supervising a 57mm engagement;

c. Supervising a CIWS engagement by:
(1) Operating combat systems (CS) equipment (B00080);
(2) Operating FC systsms and associated equipment (D00080); and
(3) Operating Anti Air Warfare (AAW) weapons and sensors (D00060);

d. Advising AWWD on functionality of the Fire Control System by:
(1) Supervising FC equipment operator functional checks (D0090); and
(2) Directing FC team in fault reconciliation.

EO 002.01 – Supervise ESSM Engagement

002.01.01 – ESSM Air Engagement

Conditions		Sequence
SEACOT (preferred) SEAWOLF/SETT (secondary option)		After PC 001
Time	**Classification**	**Instructor**
100 min	Up to SECRET	RQ MS Instructor RQ PO2 Instructor (or above)

Learning Objectives	Method
1) Supervise ESSM Air Engagement 2) CEROS display set up	EDI

Activity Steps	
	1. Instructor to demonstrate the CEROS display set up and role of the FC Sup during an ESSM Air engagement IAW NCPM 1000. Instructor to indicate key events that require specific FC Sup action/supervision. Special notes to consider: (30 min) a. Indicators on TV view b. Errors c. CWI status d. Missile status 2. Students to pair up and practice conducting the role of the FC Sup during and ESSM Air engagement with partner acting as FCO, 5 min per run, twice. Instructor to circulate throughout and provide feedback. (70 min)

Resources	References
1) NCPM 1000 x 12 (1 per student)	1) NCPM 1000

Notes to Instructor
If instructing in SEAWOLF students will need to pair up and take turns as there are limited systems to work on. May need to allocate double the scheduled time due to resource management. During instructional time, 2 students can act as FC Sup at a time with one instructor filling role of SWC and the other providing feedback on student performance. Peer support for FCO role.

EO 002.01 – Supervise ESSM Engagement

002.01.02 – ESSM Misfire

Conditions		Sequence
SEACOT (preferred) SEAWOLF/SETT (secondary option)		After EO 002.01A
Time	**Classification**	**Instructor**
100 min	Up to SECRET	RQ MS Instructor RQ PO2 Instructor (or above)

Learning Objectives	Method
1) Supervise ESSM misfire 2) Response to CWI failure during ESSM engagement 3) CEROS manipulation post misfire	EDI

Activity Steps	
	1. Instructor to demonstrate the CEROS display set up and role of the FC Sup during an ESSM Air engagement with misfire IAW NCPM 1000. Instructor to indicate key events that require specific FC Sup action/supervision. Special notes to consider: (30 min) a. Indicators on TV view b. Errors c. CWI status/CWI failure d. Missile status 2. Students to pair up and practice conducting the role of the FC Sup during an ESSM Air engagement with misfires and CWI failures, with partner acting as FCO, 5 min per run, twice. Instructor to circulate throughout and provide feedback. (70 min)

Resources	References
1) NCPM 1000 (1 per student)	1) NCPM 1000 2) NAVORD 4995-0

Notes to Instructor
If instructing in SEAWOLF students will need to pair up and take turns as there are limited systems to work on. May need to allocate double the scheduled time due to resource management. During instructional time, 2 students can act as FC Sup at a time with one instructor filling role of SWC and the other providing feedback on student performance. Peer support for FCO role.

EC 002.01 – Supervise ESSM Engagement

Scope	Conditions	Sequence
EO 002.01	SEACOT (preferred) SEAWOLF/SETT (secondary option)	After EO 002.01
Time	**Classification**	**Instructor**
250 min	Up to SECRET	AWWD Instructor RQ PO2 Instructor (or above)

Scenario	ESSM Engagement
Assessment Steps	Input: Students provided a FC Operator during a ESSM engagement scenario Performance: Student must conduct their duties as FC Sup IAW checklist 002.01 for a 20min run (based on training scenario details determined by instructor ***see instructor notes for criteria) Output: IAW checklist 002.01 Feedback: immediate verbal and written IAW checklist 002.01

Resources	References
1) NCPM 1000 (1 per student)	1) NCPM 1000 2) NAVORD 4995-0 3) CFCD 106

Notes to Instructor
Assistance shall not be provided to students during the assessment. Support manning IAW NWTP vol 2. Scenario will begin with a closed-up team having completed assuming policy. Student will conduct ESSM engagements against a simulated air contacts (FBA or BQM) until all items on the checklist have been achieved. PAC scenario must include one error of each of the following categories: • Incorrect CEROS display set-up • An incorrect voice procedure conducted by FC Op • One ESSM DUD indication • One CWI failure (immediately prior to, or during, missile launch)

Limitations	Using CBT vice CMS
Impact of Failure	Feedback and additional instruction provided (when possible)

Item	EC 002.01 Performance Statement	SAT	UNSAT	Remarks
1. ESSM Air Engagement				
Ensure FC Operator has system display setup IAW NCPM 1000		☐	☐	
Ensure FC Op drill and voice procedures is conducted IAW NCPM 1000		☐	☐	
Conducts all FC Sup reports IAW NCPM 1000		☐	☐	
Reports CWI failure to SWC in a clear and timely manner		☐	☐	
2. ESSM Misfire/DUD				
Enforce FC Op drill system manipulation is conducted IAW NCPM 1000		☐	☐	
Conducts FC Sup reports IAW NCPM 1000		☐	☐	

Score: /6

Trainee must achieve 5/6 SATs on all factors in order to pass this EC.

Positive Feedback	Improvement Feedback

Overall Assessment – Circle One

Met Standard	Did Not Meet Standard
Assessor I have discussed the results of this assessment with the candidate.	**Candidate** I have been debriefed on the result of this assessment

Assessor	Name:	Date:	Signature:
Candidate	Name:	Date:	Signature:

EO 002.02 – Supervise 57mm Engagement

002.02.01 – PAC

Conditions		Sequence
SEACOT (preferred) SEAWOLF/SETT (secondary option)		After EC 002.01
Time	**Classification**	**Instructor**
300 min	Up to SECRET	RQ MS Instructor RQ PO2 Instructor (or above)

Learning Objectives	Method
1) Supervise PAC 2) CEROS display set up	EDI

Activity Steps	
	1. Instructor to demonstrate the role of the FC Sup during PAC with misfire IAW NCPM 1000. Instructor to indicate key events that require specific FC Sup action/supervision. All reports are to be conducted using the SAR format. Special notes to consider: (60 min) 2. Indicators on TV view 3. Errors 4. Gun status 5. Muzzle velocity/MET impact 6. Students to pair up and practice conducting the role of the FC Sup during PAC with partner acting as FCO, 20 min per run. Instructor to circulate throughout and provide feedback. (240 min). Students to have exposure to: a. A system setup error (ex. Radar power, gun alert states, weapon assignment, force recoil, radar allow xmit, etc.) b. PAC setup error (surface vice air, bearing/range/altitude, magazine allocation, salvo size, etc.) c. Ballistic setting error (not updated IAW met report) d. Final calculation error (azimuth, elevation) e. An incorrect voice procedure conducted by FC Op f. Breaking engagement prior to SWK direction

Resources	References
1) NCPM 1000 (1 per student)	1) NCPM 1000 2) CFCD 106 MARTI 3221

Notes to Instructor
If instructing in SEAWOLF students will need to pair up and take turns as there are limited systems to work on. May need to allocate double the scheduled time due to resource management. During instructional time, 2 students can act as FC Sup at a time with one instructor filling role of SWC and the other providing feedback on student performance. Peer support for FCO role.

EC 002.02.01 – Supervise 57mm Engagement: PAC

Scope	Conditions	Sequence
EO 002.02A	SEACOT (preferred) SEAWOLF/SETT (secondary option)	After EO 002.02A
Time	**Classification**	**Instructor**
250 min	Up to SECRET	RQ MS Instructor RQ PO2 Instructor (or above)

Assessment Steps	Input: Students provided a FC Operator during a 57mm PAC scenario. Students will be provided a MET tech report at the start of the run. Performance: Student must conduct their duties as FC Sup IAW checklist 002.02A for a 20 min run. (based on training scenario details determined by instructor ***see instructor notes for criteria) Output: IAW checklist 002.02A Feedback: immediate verbal and written IAW checklist 002.02A

Resources	References
1) MET report 2) NCPM 1000 3) FC Op scenario/script 4) Support manning IAW NWTP vol 2	1) NCPM 1000 2) NWTP vol 2 3) NAVORD 4995-0 4) CFCD 106 MARTI 3221 (U)

Notes to Instructor

Assistance shall not be provided to students during the assessment. Support manning IAW NWTP vol 2

PAC scenario must include one of each of the following categories:

- A system setup error (ex. Radar power, gun alert states, weapon assignment, force recoil, radar allow xmit, etc.)
- PAC setup error (surface vice air, bearing/range/altitude, magazine allocation, salvo size, etc.)
- Ballistic setting error (not updated IAW met report)
- Final calculation error (azimuth, elevation)
- An incorrect voice procedure conducted by FC Op

Limitations	Using CBT vice CMS
Impact of Failure	Feedback and additional instruction provided (when possible)

Item	EC 002.02A Supervise 57mm Engagement: PAC	SAT	UNSAT	Remarks
1. PAC Preparations				
Ensure FC Operator has system display setup IAW NCPM 1000		☐	☐	
Confirm/correct FC Operator system setup		☐	☐	
Confirm/correct FC Operator ballistic data		☐	☐	
2. PAC drill conduct				
Conducts QA of FC Op PAC engagement setup		☐	☐	
Enforce FC Op drill system manipulation is conduct IAW NCPM 1000		☐	☐	
Enforce FC Op drill voice procedures are conducted IAW NCPM 1000		☐	☐	
QA input of corrections by FC Op		☐	☐	
QA input of error of the day by FC Op		☐	☐	
QA calculation error of FC Op		☐	☐	
Conducts FC Sup reports IAW NCPM 1000		☐	☐	
***Adheres to 57mm safety to include (check fires, positive weapon control orders)**		☐	☐	**CRITICAL**

Score: /11

Trainee must achieve 9/11 SATs on non-critical factors **AND** all critical factors to pass this EC.

Positive Feedback	Improvement Feedback

Overall Assessment – Circle One

Met Standard	Did Not Meet Standard
Assessor I have discussed the results of this assessment with the candidate.	**Candidate** I have been debriefed on the result of this assessment

Assessor	Name:	Date:	Signature:
Candidate	Name:	Date:	Signature:

EO 002.02 – Supervise 57mm Engagement

002.02.02 – 57mm Air Primary Procedures

Conditions		Sequence
SEACOT (preferred) SEAWOLF/SETT (secondary option)		After EC002.02A
Time	**Classification**	**Instructor**
100 min	Up to SECRET	RQ MS Instructor RQ PO2 Instructor (or above)

Learning Objectives	Method
1) 57mm Air Primary procedures 2) CEROS display set up	EDI

Activity Steps	
	1. Instructor to demonstrate the CEROS display set up and role of the FC Sup during a primary 57mm AA engagement IAW NCPM 1000. Instructor to indicate key events that require specific FC Sup action/supervision. All reports are to be conducted using the SAR format. Special notes to consider: (30 min) a. Indicators on TV view b. Errors c. Gun status (* ensure round on shift tongue prior to engagement) 2. Students to pair up and practice conducting the role of the FC Sup during a primary 57mm AA engagement with partner acting as FCO, 5 min per run, twice. Instructor to circulate throughout and provide feedback. (70 min)

Resources	References
1) NCPM 1000 (1 per student)	1) NCPM 1000

Notes to Instructor
If instructing in SEAWOLF students will need to pair up and take turns as there are limited systems to work on. May need to allocate double the scheduled time due to resource management. During instructional time, 2 students can act as FC Sup at a time with one instructor filling role of SWC and the other providing feedback on student performance. Peer support for FCO role.

EO 002.02 – Supervise 57mm Engagement

002.02.03 – 57mm Surface Procedures

Conditions		Sequence
SEACOT (preferred) SEAWOLF/SETT (secondary option)		After EO 002.02B
Time	**Classification**	**Instructor**
200 min	Up to SECRET	RQ MS Instructor RQ PO2 Instructor (or above)

Learning Objectives	Method
1) 57mm Surface Primary procedures 2) 57mm Surface Secondary procedures 3) CEROS display set up	EDI

Activity Steps	
	1. Instructor to demonstrate the CEROS display set up and role of the FC Sup during a primary 57mm SU engagement IAW NCPM 1000. Instructor to indicate key events that require specific FC Sup action/supervision. All reports are to be conducted using the SAR format. Special notes to consider: (30 min) a. Indicators on TV view b. Errors c. Gun status (* ensure round on shift tongue prior to engagement) 2. Students to pair up and practice conducting the role of the FC Sup during a primary 57mm SU engagement with partner acting as FCO, 5 min per run, twice. Instructor to circulate throughout and provide feedback. (70 min) 3. Instructor to demonstrate the CEROS display set up and role of the FC Sup during a secondary 57mm SU engagement IAW NCPM 1000. Instructor to indicate key events that require specific FC Sup action/supervision. Special notes to consider: (30 min) a. Indicators on TV view b. Errors c. Gun status (* ensure round on shift tongue prior to engagement) d. FCO allocates fire 4. Students to pair up and practice conducting the role of the FC Sup during a secondary 57mm SU engagement with partner actingg as FCO, 5 min per run, twice. Instructor to circulate throughout and provide feedback. (70 min)

Resources	References
1) NCPM 1000 (1 per student)	1) NCPM 1000

Notes to Instructor

If instructing in SEAWOLF students will need to pair up and take turns as there are limited systems to work on. May need to allocate double the scheduled time due to resource management.

During instructional time, 2 students can act as FC Sup at a time with one instructor filling role of SWC and the other providing feedback on student performance. Peer support for FCO role.

EO 002.02 – Supervise 57mm Engagement

002.02.04 – Optronic Secondary

Conditions		Sequence
SEACOT (preferred) SEAWOLF/SETT (secondary option)		After EO002.02E
Time	**Classification**	**Instructor**
100 min	Up to SECRET	RQ MS Instructor RQ PO2 Instructor (or above)

Learning Objectives	Method
1) Supervise operator in Conduct of an optronic engagement using 57mm	EDI

Activity Steps	
	1. Instructor to demonstrate the CEROS display set up and role of the FC Sup during a 57mm optronic engagement event IAW NCPM 1000. Instructor to indicate key events that require specific FC Sup action/supervision. (30 min) 2. Students to pair up and practice conducting the role of the FCS during the optronic engagement, with a partner acting as FCO. Instructor to circulate throughout and provide feedback. (70 min)

Resources	References
1) NCPM 1000 (one per student)	1) NCPM 1000

Notes to Instructor
Must highlight the fidelity of this method of tracking

EO 002.02 – Supervise 57mm Engagement

002.02.05 – 57mm Warning Shots

Conditions		Sequence
SEACOT (preferred) SEAWOLF/SETT (secondary option)		After EO 002.02F
Time	**Classification**	**Instructor**
100 min	Up to SECRET	RQ MS Instructor RQ PO2 Instructor (or above)

Learning Objectives	Method
1) Supervise 57mm warning shots drill	EDI

Activity Steps	
	1. Instructor to demonstrate the CEROS display set up and role of the FC Sup during a 57mm warning shots event IAW NCPM 1000. Instructor to indicate key events that require specific FC Sup action/supervision. All reports are to be conducted using the SAR format. (30 min) 2. Students to pair up and practice conducting the role of the FCS during the warning shots event, with a partner acting as FCO. Instructor to circulate throughout and provide feedback. (70 min)

Resources	References
1) NCPM 1000 (1 per student)	1) NCPM 1000

Notes to Instructor
If instructing in SEAWOLF students will need to pair up and take turns as there are limited systems to work on. May need to allocate double the scheduled time due to resource management. During instructional time, 2 students can act as FC Sup at a time with one instructor filling role of SWC and the other providing feedback on student performance. Peer support for FCO role.

EO 002.02 – Supervise 57mm Engagement

002.02.06 – 57mm Misfire

Conditions		Sequence
SEACOT (preferred) SEAWOLF/SETT (secondary option)		After EO 002.02G
Time	**Classification**	**Instructor**
200 min	Up to SECRET	RQ MS Instructor RQ PO2 Instructor (or above)

Learning Objectives	Method
1) Supervise 57mm Misfire	EDI

Activity Steps

1. Instructor to demonstrate the CEROS display set up and role of the FC Sup during a 57mm misfire IAW NCPM 1000. Instructor to indicate key events that require specific FC Sup action/supervision. All reports are to be conducted using the SAR format. Must include: (40 min)
    a. Hot gun
    b. Cold gun
    c. Primary locked on
    d. Secondary locked on
    e. Secondary fictitious*
    f. Check load
    g. Round on breach not struck
    h. Voltage not applied to the round
    i. Failed re-fire

2. Students to pair up and practice conducting the role of the FC Sup during a 57mm misfire with partner acting as FCO. Instructor to circulate throughout and provide feedback. (160 min)

Resources	References
1) NCPM 1000 (1 per student)	1) NCPM 1000

Notes to Instructor

If instructing in SEAWOLF students will need to pair up and take turns as there are limited systems to work on. May need to allocate double the scheduled time due to resource management.
During instructional time, 2 students can act as FC Sup at a time with one instructor filling role of SWC and the other providing feedback on student performance. Peer support for FCO role.

EC 002.02B – Supervise 57mm Engagement

Scope	Conditions	Sequence
EC 002.02	SEACOT/SEAWOLF/SETT	Following EO 002.02H
Time	**Classification**	**Instructor**
600 min	Up to SECRET	RQ-MS

Scenario	Supervise 57mm Engagement EC
Assessment Steps	Input: Students provided a FC Operator during the 57mm scenarios. Students will be provided a MET tech report at the start of the run. Performance: Student must conduct their duties as FC Sup IAW checklist 002.02B for a 20 min run. (based on training scenario details determined by instructor ***see instructor notes for criteria) Output: IAW checklist 002.02A Feedback: immediate verbal and written IAW checklist 002.02A

Resources	References
1) MET report	1) NCPM 1000
2) NCPM 1000	2) NWTP vol 2
3) FC Op scenario/script	3) NAVORD 4995-0
4) Support manning IAW NWTP vol 2	4) CFCD 106 MARTI 3221 (U)

Notes to Instructor

Assistance shall not be provided to students during the assessment. Support manning IAW NWTP vol 2

Scenario will begin with a closed-up team having completed assuming policy. Student will conduct 57mm engagements in primary or secondary. Engagement types can be any from EO 002.02, but must include a minimum of one 57mm Air engagement and one 57mm surface engagement. Run will continue until all items on the checklist have been achieved.

The scenario must include one error/stoppage of each of the following categories:

- An incorrect voice procedure conducted by FC Op
- During the air engagement: cold gun misfire with successful attempt to re-fire (SWC should order secondary-fictitious drill)
- During the surface engagement: hot gun misfire with unsuccessful attempt to re-fire, FCO neglects to select “Unload” upon completion of drill (SWC should order secondary remaining locked-on drill)
- Check Load
- No Forced-Recoil
- Gun blocked

- No rounds available (empty gun mags)
- During a surface engagement: range foul based on visual through FC camera (i.e. marine mammals)

Limitations	Using CBT vice CMS
Impact of Failure	Feedback and additional instruction provided (when possible)

Item	EC 002.02B Supervise 57mm Engagement	SAT	UNSAT	Remarks
1. 57mm Drill Conduct				
Enforce FC Op drill and voice procedures are conducted IAW NCPM 1000		☐	☐	
Conducts Cold-Gun misfire procedures IAW NCPM 1000		☐	☐	
Conducts Hot-Gun misfire procedures IAW NCPM 1000		☐	☐	
Conducts Failed Re-Fire procedures IAW NCPM 1000		☐	☐	
Conducts appropriate reactions to a Check Load indication		☐	☐	
Conducts appropriate reactions to an indication of No Forced-Recoil		☐	☐	
Conducts appropriate reactions to a lack of ammunition in the gun mags		☐	☐	
Conducts appropriate reactions to an indication of Gun Blocking in effect		☐	☐	
Conducts Check Fire procedures IAW NCPM 1000		☐	☐	**CRITICAL**
Ensures Positive Control orders are adhered to at all time		☐	☐	**CRITICAL**

Score: /10

Trainee must achieve 8/10 SATs on non-critical factors **AND** all critical factors to pass this EC.

Positive Feedback	Improvement Feedback

Overall Assessment – Circle One

Met Standard	Did Not Meet Standard
Assessor I have discussed the results of this assessment with the candidate.	**Candidate** I have been debriefed on the result of this assessment

Assessor	Name:	Date:	Signature:
Candidate	Name:	Date:	Signature:

EO 002.03 – Supervise CIWS engagement

002.03.01 – CIWS AAW Manual

Conditions		Sequence
Ship/Trainer SEACOT (secondary)		After EC 002.02B
Time	**Classification**	**Instructor**
200 min (including travel)	Up to SECRET	RQ MS Instructor

Learning Objectives	Method
1) Conduct CIWS AAW drills (manual) 2) Actions in the case of CIWS misfire	EDI

Activity Steps	
	1. Instructor to demonstrate the role of the FC Sup during a CIWS AAW engagement (manual) IAW NCPM 1000. Instructor to instruct on actions to take in the case of CIWS misfire. All reports are to be conducted using the SAR format. (30 min) 2. Students to pair up and practice conducting the role of the FC Sup during a CIWS AAW engagement (manual), instructor to circulate and provide feedback. (120 min)

Resources	References
1) NCPM 1000 (1 per student)	1) NCPM 1000

Notes to Instructor
Preferred to use a CIWS, vice trainer(procedures in the trainer differ from an actual CIWS after a surface engagement to get back to EO ready you just have to hit EO standby then EO ready.But in the trainer you have to bring the system down to standby select pre-arm/arm then Air Ready then you can go to EO standby and EO ready)

EO 002.03 – Supervise CIWS engagement

002.03.02 – CIWS Surface

Conditions		Sequence
Ship/Trainer SEACOT (secondary)		After EO 002.03A
Time	**Classification**	**Instructor**
200 min including travel	Up to SECRET	RQ MS Instructor

Learning Objectives	Method
1) Supervise CIWS surface firing drills	EDI

Activity Steps

1. Instructor to demonstrate the role of the FC Sup during a CIWS surface firing event IAW NCPM 1000. Instructor to indicate key events that require specific FC Sup action/supervision. All reports are to be conducted using the SAR format. (50 min)
    a. Apply corrections
    b. Adjusting aim
    c. Fire control modes

2. Students to pair up and practice conducting the role of the FC Sup during the CIWS surface firing event, with a partner acting as FCO. Instructor to circulate throughout and provide feedback. (120 min)

Resources	References
1) NCPM 1000	1) NCPM 1000

Notes to Instructor

Preferred to use a CIWS, vice trainer(procedures in the trainer differ from an actual CIWS after a surface engagement to get back to EO ready you just have to hit EO standby then EO ready.But in the trainer you have to bring the system down to standby select pre-arm/arm then Air Ready then you can go to EO standby and EO ready)

EC 002.03 – Supervise CIWS Engagement

Scope	Conditions	Sequence
EC 002.03	SEACOT (preferred) SEAWOLF/SETT (secondary option)	Following EO 002.03B
Time	**Classification**	**Instructor**
300 min	Up to SECRET	RQ-MS

Scenario	Supervise CIWS engagement EC
Assessment Steps	Input: Students provided a RCO during a CIWS surface and air engagement scenario. Students will be provided a policy report at the start of the run. Performance: Student must conduct their duties as FC Sup IAW checklist 002.03 for a 20 min run. (based on training scenario details determined by instructor ***see instructor notes for criteria) Output: IAW checklist 002.03

Resources	References
1) Policy report 2) NCPM 1000 3) RCO/CIWS scenario/script 4) Support manning IAN NWTP vol 2	1) NCPM 1000 2) NAVORD 4995-0

Notes to Instructor

Assistance shall not be provided to students during the assessment. Support manning IAW NWTP vol 2

Scenario will begin with a closed-up RCO and will receive a policy order from the SWC/instructor. Students will participate in CIWS Surface and CIWS AA Manual Engagements.

CIWS scenario must include one error of each of the following categories:

- A system setup error (Not inputting Policy correctly)
- Adjust aim incorrectly
- System not being taken out of VETO on engagement
- An incorrect voice procedure conducted by RCO

Limitations	Using CBT vice CMS
Impact of Failure	Feedback and additional instruction provided (when possible)

Item	EC 002.03 Performance Statement	SAT	UNSAT	Remarks
1. CIWS SURFACE				
Confirm/correct RCO system setup		☐	☐	
Confirm/correct RCO has input policy correctly		☐	☐	
Enforce RCO CIWS drill and voice procedures are conducted IAW NCPM 1000		☐	☐	
Conducts FC Sup reports IAW NCPM 1000		☐	☐	
2. CIWS AAW MANUAL				
Confirm/correct RCO system setup		☐	☐	
Enforce RCO CIWS drill system manipulation is conducted IAW NCPM 1000		☐	☐	
Enforce RCO CIWS drill voice procedures are conducted IAW NCPM 1000		☐	☐	
Conducts FC Sup reports IAW NCPM 1000		☐	☐	

Score: / 8

Trainee must achieve 7/8 SATs on non-critical factors this EC.

Positive Feedback	Improvement Feedback

Overall Assessment – Circle One			
Met Standard		**Did Not Meet Standard**	
Assessor I have discussed the results of this assessment with the candidate.		**Candidate** I have been debriefed on the result of this assessment	
Assessor	Name:	Date:	Signature:
Candidate	Name:	Date:	Signature:

EO 002.04 – Advise AWWD on functionality of FC System

002.04.01 – System Troubleshooting

Conditions		Sequence
SEACOT (preferred) SEAWOLF/SETT (secondary option)		After EC 002.03
Time	**Classification**	**Instructor**
100 min	Up to SECRET	RQ MS Instructor

Learning Objectives	Method
1) Identifying System Faults 2) Reacting to System Faults 3) Liaising with technicians	EDI

Activity Steps	
	1. Instructor to create a fault in the system. They will explain to the class what the fault is, how they identify it and what steps they would take to determine fault details. (5 min) 2. Instructor will then demonstrate how to use the system to determine fault details and indicate what they would communicate to the Tech based on the information found. (15 min) 3. Students are then to use their own systems to navigate the fault details from the instructor's example. (Instructor to do a verbal walk through while students are navigating to find fault details). (15 min) 4. Instructor to ask students to pair up and have them determine fault details for the following faults: a. Continuous Wave Illuminator Degraded b. Misfire c. Total fault d. VETO e. Oil Pressure in the FC director Students to then communicate the fault details to the instructor as they would to a technician. (Instructor to walk throughout while students complete each fault). (65 min)

Resources	References
1) Fault Scenario (demo)	1) CEROS manual

Notes to Instructor
If instructing in SEAWOLF students will need to pair up and take turns as there are limited systems to work on. May need to allocate double the scheduled time due to resource management. (if unable to load faults, use a sticky note to indicate fault on screen)

EO 002.04 – Advise AWWD on functionality of FC System

002.04.02 – Advising AWWD on System Limitations

Conditions		Sequence
SEACOT (preferred) SEAWOLF/SETT (secondary option)		After 002.04A
Time	**Classification**	**Instructor**
50 min	Up to SECRET	RQ MS Instructor

Learning Objectives	Method
1) Impact of faults on exercise/operations 2) Advising AWWD WRT sequence of events	T-Chart Guided Discussion (share experience)

Activity Steps	
	1. Instructor to break students into pairs, and provide each pair with a sequence of events and two faults that have come up in the system. In pairs, students are to determine the impact of these faults and if they will limit the sections ability to conduct any serials in the sequence of events. (T-chart) (10 min) 2. Students are then to present their scenario and indicate what the impact of these faults are to the exercise/operations. (20 min) 3. Instructor to then pose to the class (20 min): a. Have you encountered faults during exercise/operations? b. How were they dealt with? c. What was the impact of the reactions (lack of reaction)? *use this as an opportunity for students to share operational experience

Resources	References
1) 6x scenarios (sequence of events and faults) 2) Flip charts and markers	1) CEROS manual

Notes to Instructor

NPTG HQ Approval -07/06/2021
Version No: 1.9
MITE Crse: 122543

EO 002.04 – Advise AWWD on functionality of FC System

002.04.03 – Determine Faults and Advise on impact

Conditions		Sequence
SEACOT (preferred) SEAWOLF/SETT (secondary option)		After 002.04B
Time	**Classification**	**Instructor**
50 min	Up to SECRET	RQ MS Instructor

Learning Objectives	Method
1) Communicate fault impact	Practical Team Quiz

Activity Steps	
	1. Student to be given a fault (verbally or through system) and must determine the fault details, indicate how they would communicate with the tech, and what information they would pass to the AWWD regarding impact of the fault to the exercise/operations. (same scenario for entire class). (10 min) 2. Students are then to pair up and create their own scenario (including a sequence of events and faults). They will create a DS solution and pass their scenario to the instructor. (20 min) 3. Instructor to then distribute each group a scenario (created by another group), where they must determine fault details and impact to exercise/operations. (10 min) 4. Instructor to provide feedback to students based on their responses to the scenario details. (10 min)

Resources	References
1) 1x scenarios (sequence of events and faults)	1) CEROS manual

Notes to Instructor

EC 002.04 – Advise AWWD on functionality of FC System

Scope	Conditions	Sequence
EC 002.04	SEACOT (preferred) SEAWOLF/SETT (secondary)	After 002.04C
Time	**Classification**	**Instructor**
250 min	Up to SECRET	RQ MS

Scenario	EC on fault finding utilizing CEROS
Assessment Steps	Input: Students will be provided with support manning IAW NWTP vol2 leading up to a 57mm engagement. Performance: Students must conduct their duties as FC Sup IAW checklist 002.04 for a 20min run (based on training scenario details determined by instructor *** see instructor notes for criteria). Output: IAW checklist 002.04 Feedback: Immediate verbal and written IAW checklist 002.04

Resources	References
1) Fault Cards	1) NCPM 1000 2) NAVORD 4995-0 3) NWTP vol 2

Notes to Instructor

Assistance shall not be provided to students during the assessment.
Scenario will begin with a closed-up team having completed assuming policy. Prior to the engagements, several major faults shall be encountered. Scenario must include 2 major system faults injected into system leading up to a 57mm/ESSM engagement of an unmanned drone.

- Continuous Wave Illuminator Degraded
- Total fault
- Loss of forward *console*
- Loss of either FCR *cabinet*
- RCP Inoperative
- Software node failure

Objective is for students to identify the issue and communicate the fault details and how it will impact the engagement to the instructor as they would a technician and/or SWC.

Limitations	Using CBT vice CMS
Impact of Failure	Feedback and additional instruction provided (when possible)

Item	EC 002.04 Performance Statement	SAT	UNSAT	Remarks
1. Preparations				
Ensure FC Operator has system display setup IAW NCPM 1000		☐	☐	
Confirm/correct FC operator system setup		☐	☐	
2. 57mm/ESSM Engagement				
Conducts fault finding on CWI degraded status		☐	☐	
Recognize 57mm is Vetoed before engagement		☐	☐	
Conducts fault finding on check load indication		☐	☐	
Successfully communicates all faults to SWC/ TOW and its impact		☐	☐	

Score: / 6

Trainee must achieve 5/6 SATs on non-critical factors **AND** all critical factors to pass this EC.

Positive Feedback	Improvement Feedback

Overall Assessment – Circle One			
Met Standard		**Did Not Meet Standard**	
Assessor I have discussed the results of this assessment with the candidate.		**Candidate** I have been debriefed on the result of this assessment	
Assessor	Name:	Date:	Signature:
Candidate	Name:	Date:	Signature:

PO 002 – Supervise Fire Control Drills

PO 002 – Consolidation

Conditions		Sequence
SEACOT (preferred) SEAWOLF/SETT (secondary option)		
Time	**Classification**	**Instructor**
400 min	Up to SECRET	RQ MS Instructor

Learning Objectives	Method
1) Functionality of FC system	EDI

Activity Steps	
	1. Students will supervise NCPM 1000 drills (ESSM, 57mm and CIWS). a. ESSM misfire b. 57mm PAC c. 57 misfire (fictitious target) d. 57mm warning shots e. CIWS AAW manual 2. Then Instructor is to create a fault in the system. Student will explain what the fault is, how they identify it and what steps they would take to determine fault details. (30 min total) with debrief using the PC 002 checklist

Resources	References
1) Fault scenario	1) NCPM 1000

Notes to Instructor
If instructing in SEAWOLF students will need to pair up and take turns as there are limited systems to work on. May need to allocate double the scheduled time due to resource management.

PC 002 – Supervise FC Drills

Scope	Conditions	Sequence
PC 002	SEACOT (preferred) SEAWOLF/SETT (secondary option)	After PO002
Time	**Classification**	**Instructor**
600 min	Up to SECRET	RQ-PO2 NESOP RQ-MS NESOP

Scenario	Air and Surface engagement
Assessment Steps	Input: Students will be provided a FC Operator leading up to an Air engagement utilizing ESSM, 57mm and CIWS. Performance: Students must conduct their duties as FC Sup IAW checklist 002 for a 50min run (based on training scenario details determined by instructor *** see instructor notes for criteria). Output: Trainee must achieve 32 / 33 SATs on non-critical factors AND all critical factors to pass this PC. Feedback: Immediate verbal and written IAW checklist 002

Resources	References
1) PC 002 scenario 2) Support manning IAW NWTP vol 2	1) NCPM 1000 2) NWTP vol 2 3) NAVORD 4995-0 4) CFCD 106

Notes to Instructor
Assistance shall not be provided to students during the assessment. Support manning IAW NWTP vol 2 Scenario shall include a deliberate PAC, inbound air contact and surface contact. Air contact will be engaged with ESSM, 57mm and CIWS. They're will be a misfire on 57mm and DUD indication (need in checklist) on ESSM. After conclusion of air engagement, FCS to clear misfire (techs to mount)on 57mm and carry on with the surface engagement using 57 then CIWS.

Limitations	Using CBT vice CMS
Impact of Failure	Feedback and additional instruction provided (when possible)

Item	PC 002 Performance Statement	SAT	UNSAT	Remarks
1. Preparation				
Ensure FC Operator has system display setup IAW NCPM 1000		☐	☐	
Confirm/correct FC operator system/setup		☐	☐	
Confirm/correct RCO system setup		☐	☐	
Confirm/correct RCO has input policy correctly		☐	☐	
2. ESSM air engagement				
Conducts QA of FC Op ESSM engagement setup		☐	☐	
Enforce FC Op drill system manipulation is conducted IAW NCPM 1000		☐	☐	
Enforce FC Op drill voice procedures is conducted IAW NCPM 1000		☐	☐	
Conducts FC Sup reports IAW NCPM 1000		☐	☐	
***Adheres to ESSM safety to include (check fires, positive weapons control)**		☐	☐	**CRITICAL**
3. PAC Preparations				
Ensure FC Operator has system display setup IAW NCPM 1000		☐	☐	
Confirm/correct FC Operator system setup		☐	☐	
Confirm/correct FC Operator ballistic data		☐	☐	
1. PAC drill conduct				
Conducts QA of FC Op PAC engagement setup		☐	☐	
Enforce FC Op drill system manipulation is conduct IAW NCPM 1000		☐	☐	
Enforce FC Op drill voice procedures are conducted IAW NCPM 1000		☐	☐	
QA input of corrections by FC Op		☐	☐	
QA input of error of the day by FC Op		☐	☐	
QA calculation error of FC Op		☐	☐	
Conducts FC Sup reports IAW NCPM 1000		☐	☐	
***Adheres to 57mm safety to include (check fires, positive weapon control orders)**		☐	☐	**CRITICAL**

4. 57mm air engagement			
Enforce FC Op drill system manipulation is conduct IAW NCPM 1000	☐	☐	
Enforce FC Op drill voice procedures are conducted IAW NCPM 1000	☐	☐	
Conducts FC Sup reports IAW NCPM 1000	☐	☐	
***Adheres to 57mm safety to include (check fires, positive weapons control)**	☐	☐	**CRITICAL**
5. CIWS air engagement			
Enforce RCO CIWS drill system manipulation is conducted IAW NCPM 1000	☐	☐	
Enforce RCO CIWS drill voice procedures are conducted IAW NCPM 1000	☐	☐	
Conducts FC Sup reports IAW NCPM 1000	☐	☐	
***Adheres to CIWS safety to include (check fires, positive weapon control orders)**	☐	☐	**CRITICAL**
6. 57mm misfire			
Enforce FC Op drill system manipulation is conduct IAW NCPM 1000	☐	☐	
Enforce FC Op drill voice procedures are conducted IAW NCPM 1000	☐	☐	
Conducts FC Sup reports IAW NCPM 1000	☐	☐	
***Adheres to 57mm safety to include (check fires, positive weapons control)**	☐	☐	**CRITICAL**
7. 57mm surface engagement			
Enforce FC Op drill system manipulation is conduct IAW NCPM 1000	☐	☐	
Enforce FC Op drill voice procedures are conducted IAW NCPM 1000	☐	☐	
Conducts FC Sup reports IAW NCPM 1000	☐	☐	
***Adheres to 57mm safety to include (check fires, positive weapons control)**	☐	☐	**CRITICAL**
5. CIWS surface engagement			
Enforce RCO CIWS drill system manipulation is conducted IAW NCPM 1000	☐	☐	

Enforce RCO CIWS drill voice procedures are conducted IAW NCPM 1000	☐	☐	
Conducts FC Sup reports IAW NCPM 1000	☐	☐	
***Adheres to CIWS safety to include (check fires, positive weapon control orders)**	☐	☐	**CRITICAL**

Score: / 33 /7 CRITICAL

Trainee must achieve 32 / 33 SATs on non-critical factors **AND** all critical factors to pass this PC.

Positive Feedback	Improvement Feedback

Overall Assessment – Circle One			
Met Standard		**Did Not Meet Standard**	
Assessor I have discussed the results of this assessment with the candidate.		**Candidate** I have been debriefed on the result of this assessment	
Assessor	Name:	Date:	Signature:
Candidate	Name:	Date:	Signature:

PROTECTED B (when complete)

CHAPTER 3: ASSESSMENT OF TRAINEE

GENERAL

1. All Performance Checks (PCs) are based on POs in Chapter 2 of this QSP. Trainees must achieve the standard to be awarded the qualification. PCs will be administered in the form of PRACTICAL or WRITTEN assessment and the trainee must pass all POs to successfully complete this course.

ENABLING AND PERFORMANCE CHECKS

2. Enabling Checks (ECs). ECs are formative assessments that help the trainee and the instructors recognize progress or gaps in learning, so that confirmation or corrective action (such as additional practice or remedial instruction) can be provided. ECs also reinforce learning and increase the retention of material for longer periods in preparation for the PC. Each EO must have at least one EC. EC results do not factor into a trainee's successful completion of the PO.

3. Performance Checks (PCs). PCs are summative assessments used to determine if the trainee has met the standard under the conditions (where possible) of the PO. Supplemental PCs are administered upon failure of a PC and must be in the same format and have the same standard as the original PC.

RESPONSIBILITY FOR TESTING

4. Assessment of trainees is to be completed IAW the standards delineated in this QSP and IAW CFITES Volume 7 (Evaluation).

PROGRESS MONITORING

5. Monitoring learning is the responsibility of all instructional staff and supervisors. Monitoring, including unsatisfactory progress, shall adhere to CFITES Volume 6 Conduct of Instruction Programmes.

6. Continuous monitoring of trainee progress is required to provide the following:

   a. Early warning of trainee difficulties;
   b. Feedback on the effectiveness of training;
   c. Information for the Training Review Board (TRB) (if required); and
   d. Information to assist in the completion of the CF 377 Course Report.

7. Trainees experiencing difficulty in any area of performance are to be counselled with regard to the nature of their shortcomings and suggested corrective action IAW Training Establishment (TE) SOs or SOPs.

8. PC failures and supplemental PC failures are to be handled IAW TE SOs and SOPs. Supplemental ECs and PCs shall not be given no less than 24 hours of the trainee being informed of their EC or PC failure.

TRAINING REVIEW BOARD

9. The aim of the TRB process is to review a trainee's suitability for continued training and education, while ensuring that the trainee is afforded a fair and just assessment of all relevant information before a decision is made. TRBs shall be convened and conducted IAW NPTORD 4500-54 Training Review Board Process.

COURSE REPORTS

10. Course reports (CF 377) shall be prepared and distributed IAW NPTORD 4500-51.

11. Trainees will be graded pass or fail. NPTORD 4500-51 Course Reporting and Certificates, provides direction and guidelines for Letter Grading and Ranking.

12. Narratives will indicate trainee's strengths and weaknesses based on PC results to provide information to the Career Manager. Training limitations and/or incomplete training must also be included in the course report. A record of trainee counselling is to be maintained by the TE.

TASK LIST

Task Number	Task by Duty Area	T/NT	PO
	Duty Area A - COMMUNICATIONS		
A00020	Perform communication checks	**NT**	**-**
A00030	Configure communication terminal for operation	**NT**	**-**
A00060	Communicate using tactical voice procedures	**NT**	**-**
	Duty Area B - OPERATIONS		
B00080	Operate combat systems (CS) equipment	**T**	**001/002**
B00170	Perform pre-sail checks on equipment	**NT**	**-**
	Duty Area C - ELECTRONIC WARFARE (EW)		
C00020	Conduct Electronic Attacks	**NT**	**-**
C00050	Assess jamming	**T**	**001**
C0070	Carryout Electronic Protect (EP) measures	**T**	**001**
C00160	Interpret Emission Control (EMCON) policies	**NT**	**-**
C00450	Perform Area of Interest (AOI) monitoring	**NT**	**-**
C00480	Perform Electronic Attack (EA) decoys loading procedures	**T**	**001**
C00540	Perform threat analysis	**NT**	**-**
C00670	Track High Interest Vessels	**NT**	**-**
	Duty Area D - FIRE CONTROL		
D00010	Operate Anti Surface Warfare (ASuW) weapons and sensors	**NT**	**-**
D00020	Prepare surface targets for gunnery exercises	**NT**	**-**
D00040	Supervise Fire Control (FC) systems operation	**T**	**001/002**
D00060	Operate Anti Air Warfare (AAW) weapons and sensors	**T**	**002**
D00080	Operate FC systems and associated equipment	**T**	**002**
D00090	Supervise FC equipment operator functional checks	**T**	**002**
D00110	Supervise FC system tracking and assessments	**T**	**001**
D00130	Supervise FC systems operation	**T**	**001/002**
	Duty Area E - OPERATIONAL & ANALYSIS CENTRES		
F00080	Conduct/Participate in technical evaluations, tests and trials	**NT**	**-**
F00180	Identify Maritime, Air, and Land based platforms	**NT**	**-**
	Duty Area F - TRAINING AND EDUCATION		
G00020	Conduct individual training events	**NT**	**-**
G00040	Operate simulator/trainer systems	**NT**	**-**
G00070	Assess individual training performance	**NT**	**-**
G00090	Conduct team training events	**NT**	**-**
G00140	Assess team training performance	**NT**	**-**

RECORD OF DECISIONS

Dates:

Location:

Board Members:

Surname	Initials	Rank	MOSID	Unit
Liebersbach	A.	Lt(N)	00207	NTDC(A)
Murray	M.	PO1	00115	NTDC(A)
Parsons	S.	PO2	00115	NTDC(A)
Renaud	E.	MS	00115	NTDC(A)
Carr	D.	MS	00115	NTDC(A)
MacArthur	D.	MS	00115	NTDC(A)
Jawornicki	B.	SLt	00211	NTDC(A)

Working Group Objective:

The objective of this working group is to conduct updates to the RQ-S1 NESOP QSP. Issues were noted when the Combat Training Review was conducted Jan-Feb 2019 where the recommendation to conduct a QSP review board was recommended. As a result, the QSP was updated to reflect the changes identified and input to the FireFly format.

Specification Task

No Train Decisions:

Task #	Task Statement	Substantiation
A00020	Perform communication checks	Taught on RQ-S3 course
A00030	Configure communication terminal for operation	Taught on RQ-S3 course
A00060	Communicate using tactical voice procedures	Taught on RQ-S3 course
B00170	Perform pre-sail checks on equipment	Currently a tech job
C00020	Conduct electronic attacks	Taught on RQ-PO2 course
C00450	Perform area of interest (AOI)monitoring	Taught on RQ-OS course
C00540	Perform threat analysis	Taught on RQ-PO2 course
C00670	Track high interest vessels	Taught on RQ-S3 course
D00020	Prepare surface targets for gunnery exercises	Taught on RQ-S2 course
D00010	Operate Anti Surfacde Warfare (ASuW) weapons and sensors	Not conducted at the MS rank
F00080	Conduct participate in technical evaluations, tests and trials	Taught trough OJT
F00180	Identify maritime, air, and land based platforms	Taught trough OJT
G00020	Conduct individual training events	Taught trough OJT
G00040	Operate simulator trainer systems	Taught on RQ-S1 course
G00070	Assess individual training performance	Taught trough OJT

G00090	Conduct team training events	Taught trough OJT
G00140	Assess team training performance	Taught trough OJT

Discussions / Decisions:

1. The working group reviewed the old QSP. The consensus was that several changes were required to be made in order to modernized the course and aligned its content with FC Sup role and responsibilities post FELEX. Based on this fact, the working group came to the decision that the major milestones of the course should be as follows;
   a) Lead the FC team in prepration
   b) Supervising FC team
   c) Torubleshooting FC systems

2. The board discussed the FC systems and the fact that the old QSP did not include system faults or standard layout for the CEROS System . The working group concluded that making these changes would be a high priority as the FC Sup are currently utilizing these systems in their day to day operations.

3. The board decided that the current practice of instructing this course using visual aids such as PowerPoint is inefficient and that the students will spend most of this portion in a synthetic trainer. Therefore the focus of this QSP allowed for an increase in hands on equipment opportunities.

Guidance to Development and/or Instructional Staff:

1. The Alternate Instructional Techniques Learner Manual (change 2) was used as a resource for the development of the lesson guidance provided. Development and instructional staff should refer to the AIT Learner Manual for best practices of the delivery of lessons using alternate instructional techniques.

2. The Scheduling Annex is a draft annex that was developed in consultation with the Campus Atlantic Scheduling Section during the conduct of this working group. Lessons learned /issues with course scheduling completed using this annex should be brought to the attention of LSC staff.

Follow Up Action:

Item	Action	OPI	Expected Date of Completion
1	Set-up Learing Portal Group	NTDC(A) – CCD Analysts	

COSTING INFORMATION

Calculations are based on 6 students being from out of area on TD

Course: RQ MS NESOP	Serial:					
TD	No. Of Students / instructors /	Cost per each	# Days (if applicable)	Initial TE Set-up	Each successive course	5 Year total
Airfare (Return)	6	$1,600.00		$9,600	$9,600	$96,000.00
R&Q	6	$78.75	17	$8,032.50	$8,032.50	$80,325.00
Incidentals	6	$17.30	17	$1,764.60	$1,764.60	$17,646.00
Subtotal				**$19,397.10**	**$19,397.10**	**$197,397.10**
Material	Quantity	Cost per each	# Days (enter 1 if not applicable)			
PP&S (Estimate)	10	$50.00	17	$5000	$5000	$50,000.00
Classroom (square meters)	30	$13.00	17	…	…	…
Computer requirements	12	$0.00	17	$0	$0	$0.00
IT Equip (Proj, Smart board etc)	3	$0.00	17	$0	$0	$0.00
Instructional Materials (Printed / Handouts etc)	12	$30.00	17	$510.00	$510.00	$5,100.00
Subtotal				**$5,510.00**	**$5,510.00**	**$55,100.00**
PYs (only if course includes incremental on SWE, or course to be delivered off site	Quantity	Cost per day* (SWE if necessary)	# Days (if applicable)			
Instructor (PO1) (incremental)5	0	$0.00	0	$0	$0	$0.00
Instructor (Lt(N)) (incremental)5	0	$0.00	0	$0	$0	$0.00
Incremental Civilian Instructor	0	0	0	$0	$0	$0.00
		subtotal		**$0.00**	**$0.00**	**$0.00**
	Total Estimated Costs			**$24,907.10**	**$24,907.10**	**$249,071.00**
Course: RQ-NESOP-MS	Serial:					

ASSESSMENT PLAN

EC/PC	Scope Covered	Purpose	How	Pass/Fail Criteria
EC 001.01	EO 001.0.01 EO 001.01.02	To assess student's understanding of Pre-firing Preprations	Verbal/Practical	5/6 SATs IAW EC 001.01 Checklist
EC 001.02	EO 001.02.01 EO 001.02.02	To assess student's ability to optimize FC settings	Practical	5/6 SATs IAW EC 001.02 Checklist
PC 001	PO 001	To assess the student's ability to prepare FC systems for a successful shoot	Practical	10/12 SATs IAW PC 001 Checklist
EC 002.01	EO 002.01.01 EO 002.01.02	To assess the student's ability to supervise an ESSM engagement	Practical	5/6 SATs IAW EC 002.01 Checklist
EC 002.02.01	EO 002.02.01	To assess the student's ability to supervise an 57mm PAC engagement	Practical	9/11 SATs IAW EC 002.02A Checklist
EC 002.02.02	EO 002.02.02 EO 002.02.03 EO 002.02.04 EO 002.02.05 EO 002.02.06	To assess the student's ability to supervise an 57mm engagement	Practical	8/10 SATs IAW PC 002.02B Checklist
EC 002.03	EO 002.03.01 EO 002.03.02	To assess the student's ability to supervise an CIWS engagement	Practical	8/10 SATs IAW EC 002.03 Checklist
002.04	EO 002.04.01 EO 002.04.02 EO 002.04.03	To assess the student's FC system knowledge and troubleshooting ability	Practical	5/6 SATs IAW EC 002.04 Checklist
PC 002	PO 002	To assess the student's ability to supervise FC system	Practical	32/40 SATs IAW PC 002 Checklist

SCHEDULING INFORMATION

Scheduling Information- RQ_MS NESOP													
Lesson		Location	IT Assets					Footprint				Instructor	
#	Time	Location	DWAN	CSNI	SSA	Proj	TEL	Sec Class	WB	Desks	Type	Qual	Sec Cl
RESIDENTIAL PHASE													
EO 001.01	100	Class room	y	n	12	y		Unclass	y	12	MIL	RQ MS	SEC
EO 001.01.02	100	Class room	y	n	12	y		Unclass	y	12	MIL	RQ MS	
EC 001.01	150	Class room	y	n	12	n		Unclass	y	12	MIL	RQ MS	SEC
EO 001.02.01	100	Class room	y	n	12	y		Unclass	y	12	MIL	RQ MS or guest from CFMWC	
EO 001.02.02	50	SEA COT / SEA WO LF	n	n	12	y		SEC	y	12	MIL	RQ MS	SEC
EC 002.01	250	SEA COT / SEA WO LF	n	n	12	y		SEC	y	12	MIL	RQ MS	SEC
EO 002.02.01	300	SEA COT /	n	n	12	y		SEC	y	12	MIL	RQ MS	

		SEA WO LF											
EC 002.02.01	250	SEA COT /SEA WO LF	n	n	12	y		SEC	y	12	MIL	RQ MS	SEC
EO 002.02.01	100	SEA COT /SEA WO LF	n	n	12	y		SEC	y	12	MIL	RQ MS	
EO 002.02.03	200	SEA COT / SEA WO LF	n	n	12	y		SEC	y	12	MIL	RQ MS	SEC
EO 002.02.04	100	SEA COT /SEA WO LF	n	n	12	y		SEC	y	12	MIL	RQ MS	
EO 002.02.05	100	SEA COT /SEA WO LF	n	n	12	y		SEC	y	12	MIL	RQ MS	
EO 002.02.06	200	SEA COT / SEA WO LF	n	n	12	y		SEC	y	12	MIL	RQ MS	SEC

EC 002.02.02	600	SEA COT / SEA WO LF	n	n	12	y		SEC	y	12	MIL	RQ MS	
EO 002.03.01	200	SEA COT / SEA WO LF	n	n	12	y		SEC	y	12	MIL	RQ MS	SEC
EO 002.03.02	200	SEA COT / SEA WO LF	n	n	12	y		SEC	y	12	MIL	RQ MS	SEC
EC 002.03	300	SEA COT /SEA WO LF	n	n	12	y		SEC	y	12	MIL	RQ MS	SEC
EO 002.04.01	100	SEA COT /SEA WO LF	n	n	12	y		SEC	y	12	MIL	RQ MS	SEC
EO 002.04.02	50	SEA COT /SEA WO LF	n	n	12	y		SEC	y	12	MIL	RQ MS	SEC
EO 002.04.03	50	SEA COT /SEA	n	n	12	y		SEC	y	12	MIL	RQ MS	SEC

		WO LF											
EC 002.04	250	SEA COT /SEA WO LF	n	n	12	y		SEC	y	12	MIL	RQ MS	SEC
PO 002 Consolidtion	400	SEA COT /SEA WO LF	n	n	12	y		SEC	y	12	MIL	RQ MS	SEC
PC 002	600	SEA COT /SEA WO LF	n	n	12	y		SEC	y	12	MIL	RQ MS	SEC

Recommended Schedule

Time	Day 1	Day 2	Day3	Day 4	Day 5
Period 1 0800-0850	Expectations /Admin ISSO ,IT, Command Intro	001.01.02 Supervise MASS loading	001.02.02 Confirm System Settings	PC 001 Lead FC Team Preps	EC 002.01 Supervise ESSM
Period 2 0850-0940			EC 001.02 Optimize FC settings		
0940-0950	**Break**				
Period 3 0950-1040	**BAT**	EC 001.01 Supervise Pre Firing PREPS	EC 001.02 Optimize FC settings	PC 001 Lead FC Team Preps	EC 002.01 Supervise ESSM
Period 4 1040-1130			**PT**	**PT**	
1130-1230	**Lunch**				
Period 5 1230-1320	**TAT**	EC 001.01 Supervise Pre Firing PREPS	EC 001.02 Optimize FC settings	002.01.01 ESSM Air	EC 002.01 Supervise ESSM
Period 6 1320-1410					002.02.01 PAC
1410-1420	**Break**				
Period 7 1420-1510	001.01.01 Supervise Pre-Firing Preps	001.02.01 PAC Theory	PC 001 Lead FC Team Preps	002.02.02 ESSM Misfire	002.02.01 PAC
Period 8 1510-1600					PT

Recommended Schedule

Time	Day 6	Day 7	Day 8	Day 9	Day 10
Period 1 0800-0850	002.02.01 PAC	EC 002.02.01 Supervise PAC	002.02.04 Optronic Secondary	002.02.06 57mm Misfire	EC 002.02B Supervise 57mm Engagements
Period 2 0850-0940					
0940-0950					
Period 3 0950-1040	002.02.01 PAC	002.02.02 57mm Air primary	**PT**	EC 002.02B Supervise 57mm Engagements	EC 002.02B Supervise 57mm Engagements
Period 4 1040-1130				**PT**	**TAT**
1130-1230					
Period 5 1230-1320	**BAT**	002.02.03 57mm Surface	002.02.05 57mm earning Shots	EC 002.02B Supervise 57mm Engagements	EC 002.02B Supervise 57mm Engagements
Period 6 1320-1410	EC 002.02.01 Supervise PAC				
1410-1420					
Period 7 1420-1510	EC 002.02.01 Supervise PAC	002.02.03 57mm Surface	002.02.06 57mm Misfire	EC 002.02B Supervise 57mm Engagements	EC 002.02B Supervise 57mm Engagements
Period 8 1510-1600					

Recommended Schedule

Time	Day 11	Day 12	Day 13	Day 14	Day 15
Period 1 0800-0850	002.03.01 CIWS AWW Manual	EC 002.03 Supervise CIWS	002.04.01 System Troubleshooting	EC 002.04 Advise AWWD	PO 002 Consolidation
Period 2 0850-0940				PO 002 Consolidation	
0940-0950					
Period 3 0950-1040	002.03.01 CIWS AWW Manual	EC 002.03.02 CIWS Surface	002.04.02 Advising The AWWD	PO 002 Consolidation	PC 002 Supervise FC Drills
Period 4 1040-1130			**PT**	**PT**	**TAT**
1130-1230					
Period 5 1230-1320	002.03.02 CIWS Surface	EC 002.03.02 CIWS Surface	002.04.03 Determine Faults	PO 002 Consolidation	PC 002 Supervise FC Drills
Period 6 1320-1410			EC 002.04 Advise AWWD		
1410-1420					
Period 7 1420-1510	002.03.02 CIWS Surface	**PT**	EC 002.04 Advise AWWD	PO 002 Consolidation	PC 002 Supervise FC Drills
Period 8 1510-1600		**BAT**			

Recommended Schedule

Time	Day 16	Day 17			
Period 1 0800-0850	PC 002 Supervise FC Drills	**TAT Grad**			
Period 2 0850-0940					
0940-0950					
Period 3 0950-1040	PC 002 Supervise FC Drills	**BAT PCD**			
Period 4 1040-1130					
1130-1230					
Period 5 1230-1320	PC 002 Supervise FC Drills				
Period 6 1320-1410					
1410-1420					
Period 7 1420-1510	PC 002 Supervise FC Drills				
Period 8 1510-1600	**PT**				

TEL DEVELOPMENT

Title	Type	Description of Requirement	Lesson	SOR #
MASS loading video	Video	A video of MASS loaders in incorrect dress and other improper ammunition handling.	EO001.01B	
Track assessment	Video	A video showing the CEROS`s A-scope during a SAT UNSAT tracking assessment	EO001.01A	
MASS effect on tracking	Video	A video showing the effect a MASS cloud has on a fire control radar	EO001.01A	

REFERENCES

Ref	Abbreviated Title	Long Version Title
1	CEROS Manual	CEROS Manual
2	CFCD 106	Canadian Forces Command Douctrine 106
3	NAVORD 4995-0	Naval Orders 4995-0
4	NCPM 1000	Naval Combat Procedures Manual 1000
5	NWTP vol 2	Naval Warfare Training Publications vol 2
6		
7		
8		

PLAN, TRAIN, DO

PO	EO	Maintain Readiness	Plan	Train	Execute
001	01				Supervise Pre-firing preps
	02				Optimizing FC settings
002	01				Supervise ESSM Engagements
	02			`	Supervise 57mm engagements
	03				Supervise CIWS Engagements
	04				Advuse AWWD on Fynctionally of FC systtem